Beyond BMI: Predicting Metabolic Syndrome with Gender-Aware Feature Engineering

Ria Raheja
Barnard College
New York, NY
rr3499@barnard.edu

Sachi Patel
Barnard College
New York, NY
sdp2173@barnard.edu

Diya Nair
Barnard College
New York, NY
dn2549@barnard.edu

***Abstract*—Body Mass Index (BMI) is widely used as a clinical screening tool but does not account for differences in fat distribution, sex-specific risk gradients, or population-level variation, which can limit its accuracy as a predictor of cardiometabolic risk. We evaluate whether a small set of engineered anthropometric and demographic predictors improves prediction of metabolic syndrome (MetS) beyond BMI alone. Using data from the National Health and Nutrition Examination Survey (NHANES), we compare a baseline screening-style model (BMI, age, sex) to an enhanced feature set that adds waist-to-height ratio (WHtR), a BMI $\times$ sex interaction term, and race/ethnicity indicators. Both feature sets are evaluated across elastic net logistic regression, random forest, and LightGBM classifiers to isolate whether performance gains arise from improved feature representation rather than model complexity. The enhanced feature set significantly improves AUROC across all three classifiers ($p < 0.001$), with the best-performing model (enhanced elastic net logistic regression, AUROC = 0.812) outperforming all baseline models. SHAP analysis confirms that WHtR and the BMI $\times$ sex interaction are the primary drivers of improvement.**

***Index Terms*—BMI, metabolic syndrome, feature engineering, gender-aware models, NHANES, machine learning**

I. Introduction

Body Mass Index (BMI) is a commonly-used health metric used to screen for macro-nutritional status, including undernutrition, normal weight, and over-nutrition [1], [2]. It was developed in the 1830s by Adolphe Quetelet, a Belgian mathematician, to allow for the comparison of White, middle-aged European men of varying heights. It was originally known as the Quetelet index [1], [2]. The measure was later termed "the body mass index" by Ancel Keys in 1972, or BMI [1]. BMI is calculated by dividing an individual's weight in kilograms by the square of their height in meters (kg/m^2) and applying the result to set ranges [1]. Standard clinical categories identify four main ranges: underweight ($<$18.5 kg/m^2), normal (18.5–$<$25.0 kg/m^2), overweight (25.0–$<$30.0 kg/m^2), and obese ($\geq$30.0 kg/m^2) [1], [3].

BMI is primarily used as a rapid clinical screening tool, in population health surveys to identify trends in weight, and to estimate risks for chronic conditions like cardiovascular disease, diabetes, and certain cancers [1], [2]. It is beneficial because it is simple, inexpensive, and accessible, requiring only weight and height measurements that can be easily obtained by healthcare workers or individuals at home without the need for radiation, imaging, or complex laboratory tests [1], [2]. Additionally, its long-standing history provides a familiar benchmark for comparing contemporary health data with historical trends across several decades [2].

BMI can be considered problematic for a few different reasons. First, BMI can be problematic because it fails to differentiate between muscle mass, bone density, and fat [1], [2]. For example, BMI can often lead to the misclassifications of muscular athletes as overweight or obese, while potentially missing excess fat in older adults who may fall within a "normal" range [1], [2]. Additionally, BMI does not measure fat distribution, such as harmful abdominal or visceral fat, which is a better predictor of mortality and metabolic disease than total weight [1], [3]. Furthermore, its reliance on a historical sample of White European men makes it racially and ethnically biased, as different ethnic groups may have varying proportions of muscle and fat at the same BMI, leading to inaccurate risk assessments for diverse populations and an incorrect assumption that BMI is universally applicable [1], [2].

Therefore, we propose that a more gender-aware, ethnicity-aware model using engineered features (BMI $\times$ sex interaction, waist-to-height ratio, body fat estimates) will significantly outperform raw BMI in predicting metabolic syndrome or cardiovascular risk. We test this hypothesis using NHANES data, comparing baseline and enhanced feature sets across three classifiers of increasing complexity, and interpret the results using SHAP analysis to confirm that performance gains are driven by clinically meaningful predictors.

II. Related Work

A. Pro-BMI Literature

In "Association of All-Cause Mortality with Overweight and Obesity Using Standard Body Mass Index Categories," Katherine M. Flegal et al. (2013) analyze data from multiple large national surveys to examine how BMI relates to mortality risk. The study finds that BMI is strongly linked to health outcomes across populations. Individuals in higher BMI categories, especially obesity, show increased risk of death. At the same time, the results highlight that BMI categories can still capture meaningful differences in risk even without more detailed body composition data. Since the study uses large and representative datasets, it shows that BMI is a reliable and consistent measure at the population level. These findings

support using BMI as a baseline feature in our model since it reflects broad patterns of disease risk and mortality. [4]

In "Body Mass Index vs Body Fat Percentage as a Predictor of Health Outcomes," Mainous et al. (2025) examine how well BMI predicts disease risk compared to more direct measures of body fat. The authors find that BMI remains widely used in clinical and public health settings because it is easy to calculate and strongly linked to risks such as diabetes and cardiovascular disease. Major health organizations such as the Centers for Disease Control and Prevention, National Institutes of Health, and World Health Organization continue to endorse BMI as a standard measure. Though the paper acknowledges that BMI can misclassify some individuals, it emphasizes that more precise methods are often expensive or harder to use in practice. BMI remains a practical and scalable tool for identifying at-risk populations. These findings support keeping BMI in our baseline model where it serves as an accessible and standardized reference point for comparing more complex features [5].

B. Anti-BMI Literature

"Body Mass Index Obesity, BMI, and Health: A Critical Review" by Frank Q. Nuttall, MD, PhD, explores historical and present social body standards and medical indexes of weight as a health indicator. The author critiques the standardization of the Body Mass Index (BMI) that is popularly used in medicine, indicating that it is not dynamic enough to be widely generalizable. The paper compares different BMI population distributions including Carnegie Institute across Adult Americans, World Health Organization BMI Quartile organization of international populations, and general western population categorization of BMI. Nuttall identifies that a key problem of using BMI is that "it does not differentiate between body lean mass and body fat mass," and thus indicating an individual as overweight when they have a large muscle mass and small body fat percentage. Other limitations of BMI include changes in height trends, body fat location, sex differences in body fat location and mass, and poor indication of health risks. This paper informs our model creation in suggestions of considerations we should include in our more comprehensive predictor. By recognizing specific limitations in the article, we can tune our model to be a more accurate predictor of health issues [3].

In "Advantages and Limitations of the Body Mass Index (BMI) to Assess Adult Obesity" by Wu, Li, and Vermund, the authors do a widespread review of literature on BMI advantages, limitations, and risk indication accuracy. Additionally, they compare and contrast historical BMI trends to current use of BMI as an indication of overweight individuals. The authors used scientific articles on BMI from Google Scholar and PubMed to analyze BMI benefits, limitations, and bias. Comparisons were also drawn to BMI literature from the U.S. Centers of Disease Control and Prevention (CDC), the National Institutes of Health (NIH), The Lancet Diabetes and Endocrinology Commission, the AMA Council on Science and Public Health critique of BMI, and the World Health Organization (WHO). The three limitations to using BMI in healthcare as a screening tool for obesity identified in the paper include: (1) higher BMI does not necessarily indicate excessive body fat, (2) simplicity of BMI does not ensure its validity for screening in all populations, and (3) BMI is not an adequate or comprehensive predictor of higher disease risk. The authors identify that supplemental information to BMI use should include nutritional evaluations, restructuring BMI cut-offs to newer information on demographics, and including other medical assessments when discussing weight-based interventions. The information compiled and analyzed in this paper highlights the limitations of the use of BMI in indicating health risks. Using our ML model to evaluate the accuracy of BMI in making health predictions, we can emphasize the significance of these limitations identified [1].

C. Alternative Measure Literature

In "Waist-to-Height Ratio as a Screening Tool for Obesity and Cardiometabolic Risk," Yoo (2016) reviews evidence from multiple epidemiological studies and meta-analyses to evaluate waist-to-height ratio (WHtR) as an alternative to BMI for identifying central obesity and cardiometabolic risk in adults. The review finds that WHtR consistently outperforms BMI in predicting cardiometabolic outcomes because it captures abdominal fat distribution rather than total body weight. WHtR also performs at least as well as waist circumference. It has a practical advantage with a simple guideline: "keep your waist circumference less than half your height." This rule does not require sex- or age-specific charts. This simplicity makes WHtR useful for both clinical and community screening. WHtR is one of the key engineered features in our proposed model. Yoo's findings support including WHtR as a stronger alternative to raw BMI. Our work builds on this by combining WHtR with gender-aware features in a machine learning framework rather than studying it alone. [6].

Behkamal et al. (2025) propose a machine learning framework for predicting metabolic syndrome (MetS) using serum liver function tests, high-sensitivity C-reactive protein, and basic demographics such as age and sex. Their data comes from the MASHAD cohort in Iran from 2010 to 2020 with about 9,704 participants. The authors compare seven classifiers including logistic regression, decision tree, SVM, random forest, balanced bagging, gradient boosting, and CNN. They use an 80/10/10 train, test, and validation split and apply SVM-SMOTE to handle class imbalance. Gradient boosting performs best with an AUROC of about 0.72 and an F1 score of about 0.72. SHAP analysis shows that hs-CRP, direct bilirubin, ALT, and sex are the most important features. This study provides a strong baseline for machine learning prediction of MetS. It also shows that sex plays an important role in metabolic risk. However, their model focuses on lab biomarkers and does not include body-based measures. It also does not account for gender-aware body composition. Our study builds on this by testing whether features like BMI $\times$ sex, WHtR, and estimated body fat can achieve similar or

better performance using more accessible data that does not require lab testing. [7].

III. Methodology

A. Data and Outcome

This study uses publicly available data from the National Health and Nutrition Examination Survey (NHANES), a nationally representative cross-sectional survey conducted by the CDC. We use the 2017–March 2020 pre-pandemic cycle. Six component tables are merged by respondent ID (SEQN): Demographics (DEMO), Body Measures (BMX), Blood Pressure (BPX), HDL Cholesterol (HDL), Triglycerides (TRIGLY), and Fasting Glucose (GLU). The sample is limited to non-pregnant adults aged 18+ with complete data across these tables, yielding an analytic sample of 3,568 participants.

The outcome variable is metabolic syndrome (MetS). We chose MetS because it captures a cluster of interconnected cardiometabolic risk factors (high blood pressure, elevated fasting glucose, excess abdominal adiposity, and abnormal lipid levels) that together increase the risk of cardiovascular disease and type 2 diabetes. This makes it a stronger test of whether BMI alone is sufficient than predicting any single condition. A participant is classified as having MetS if three or more of the following are present: elevated waist circumference ($\geq$88 cm women, $\geq$102 cm men), triglycerides $\geq$150 mg/dL, low HDL cholesterol ($<$50 mg/dL women, $<$40 mg/dL men), blood pressure $\geq$130/85 mmHg, or fasting glucose $\geq$100 mg/dL.

Because waist circumference is itself part of the MetS diagnostic criteria, we avoid including raw waist circumference as a predictor to prevent label leakage. Instead, we construct waist-to-height ratio (WHtR), which captures central adiposity while remaining interpretable as a derived anthropometric feature rather than directly duplicating a diagnostic threshold component of the outcome definition.

B. Feature Engineering

To test our hypothesis, we compare two feature sets. The baseline set uses only BMI, age, and sex (mimicking how BMI is commonly used in clinical screening and population-level risk adjustment models). The enhanced set adds waist-to-height ratio (WHtR), a BMI $\times$ sex interaction term, and race/ethnicity indicators. WHtR captures variation in abdominal fat distribution that BMI alone cannot represent. The interaction term allows the model to estimate sex-specific BMI risk gradients instead of assuming a shared slope across sexes. Race/ethnicity is included as a categorical predictor encoded using indicator (one-hot) variables, allowing the model to adjust the BMI–risk relationship across population groups without imposing a linear ordering across categories.

C. Models

We train three classifiers on each feature set: elastic net logistic regression, random forest, and LightGBM. These models were selected intentionally to represent increasing modeling flexibility while remaining well-suited to structured clinical tabular data. Elastic net logistic regression provides an interpretable linear baseline and stabilizes coefficient estimates in the presence of correlated anthropometric predictors through regularization. Random forest captures nonlinear relationships and interaction effects without requiring manual specification. LightGBM, a gradient-boosted decision tree model optimized for tabular datasets, provides a high-performance benchmark capable of learning complex feature interactions efficiently while maintaining strong predictive accuracy on structured health data. Using multiple model classes allows us to distinguish whether performance gains arise from improved feature engineering rather than from algorithm choice alone. Each feature set is therefore evaluated across all three classifiers for consistent comparison.

D. Evaluation

Models are trained and evaluated using stratified 5-fold cross-validation to preserve the prevalence of metabolic syndrome across folds and produce stable performance estimates. The primary evaluation metric is area under the receiver operating characteristic curve (AUROC), which measures discrimination between individuals with and without MetS independent of classification threshold. Secondary metrics include precision, recall (sensitivity), specificity, and F1 score to evaluate classification performance under class imbalance typical of screening-style prediction tasks. Statistical significance of AUROC differences between baseline and enhanced feature sets is assessed using a paired bootstrap test with 10,000 resamples. To interpret model behavior, SHAP (Shapley Additive Explanations) values are computed for the best-performing model.

To ensure that reported performance reflects generalizable model behavior rather than overfitting to a particular train-test split, all models are evaluated under a unified stratified 5-fold cross-validation framework. Stratification is applied at the fold level to preserve the approximately 36% MetS prevalence observed in the analytic sample across each fold, preventing artificially inflated or deflated performance estimates that could arise from imbalanced partitioning. Hyperparameter tuning for the elastic net (regularization strength α, mixing parameter λ), random forest (number of estimators, maximum depth, minimum samples per leaf), and LightGBM (learning rate, number of leaves, subsample ratio) is conducted via nested cross-validation within each training fold to prevent information leakage from the validation set into model selection decisions. Final reported metrics represent the mean and standard deviation of fold-level scores, providing both a point estimate of expected performance and a measure of variance across data partitions. All pre-processing steps, including feature scaling for the elastic net and one-hot encoding of race/ethnicity, are applied within each fold's training pipeline to avoid data leakage from the held-out validation observations.

The analytic sample exhibits moderate class imbalance, with approximately 36% of participants meeting the MetS criteria and 64% not, a distribution typical of population-based screening cohorts. While this imbalance is less severe than in many

clinical prediction tasks, it is sufficient to bias model training toward the majority class if left unaddressed, manifesting as inflated accuracy alongside suppressed sensitivity for the positive class. We do not apply resampling techniques such as SMOTE or random oversampling, as prior work has shown these methods can distort the feature covariance structure in anthropometric data and introduce synthetic observations that do not reflect plausible clinical profiles. Instead, class imbalance is addressed through two complementary choices. First, all classifiers are trained with class-weighted loss functions, which penalize misclassifications of the minority MetS-positive class in proportion to its underrepresentation, directing model optimization toward improved sensitivity without artificially inflating the training distribution. Second, AUROC is selected as the primary evaluation metric precisely because it evaluates discrimination across all classification thresholds simultaneously and is invariant to class prevalence, making it more informative than accuracy under imbalance. Recall and F1 score are reported as secondary metrics to characterize the sensitivity-precision tradeoff at the default decision threshold, and results are interpreted with awareness that threshold selection in clinical deployment should be calibrated to the specific cost asymmetry between false negatives and false positives in a screening context.

IV. Results

A. Overall Model Performance

Table I reports mean classification metrics with standard deviations across stratified 5-fold cross-validation for all six model–feature set combinations. The enhanced feature set improves AUROC over the baseline across all three classifiers: elastic net logistic regression rises from 0.791 to 0.812, random forest from 0.779 to 0.811, and LightGBM from 0.771 to 0.793. F1 score, precision, and recall follow the same pattern with the enhanced set producing consistent gains regardless of model class. The narrow standard deviations ($\leq$0.009 for AUROC) indicate stable performance across folds.

Notably, the simplest model with enhanced features (elastic net logistic regression, AUROC = 0.812) outperforms both complex models trained on baseline features alone (random forest at 0.779, LightGBM at 0.771). This indicates that the performance improvement is driven by feature representation rather than model complexity, directly supporting our hypothesis.

TABLE I
Classification performance (mean $\pm$ SD, 5-fold stratified CV).

Set	Model	AUROC	F1	Recall
Base	EN-LR	.791$\pm$.008	.538$\pm$.013	.463$\pm$.020
Base	RF	.779$\pm$.008	.591$\pm$.015	.586$\pm$.027
Base	LGBM	.771$\pm$.006	.573$\pm$.016	.562$\pm$.027
Enh	EN-LR	.812$\pm$.006	.593$\pm$.006	.535$\pm$.007
Enh	RF	.811$\pm$.009	.625$\pm$.007	.629$\pm$.006
Enh	LGBM	.793$\pm$.009	.610$\pm$.014	.607$\pm$.022

EN-LR = Elastic Net LR, RF = Random Forest,
LGBM = LightGBM. Base = BMI, age, sex.
Enh = + WHtR, BMI $\times$ sex, race/ethnicity.

B. Statistical Significance

To confirm that the AUROC improvements are not attributable to chance, we also conduct paired bootstrap tests (10,000 resamples) comparing enhanced and baseline feature sets within each model class. All three comparisons are statistically significant: elastic net logistic regression (ΔAUC = +0.021, z = 5.56, $p < 0.001$), random forest (ΔAUC = +0.032, z = 8.19, $p < 0.001$), and LightGBM (ΔAUC = +0.022, z = 4.49, $p < 0.001$). The largest absolute gain occurs for random forest, where the enhanced features close the gap between the linear and tree-based models.

C. ROC Curve Analysis

Fig. 1 displays ROC curves for all six model–feature set combinations, organized by classifier class with baseline (blue) and enhanced (orange) curves plotted together in each panel. In every panel, the enhanced curve lies above the baseline curve across the full range of false positive rates, confirming that the performance improvement from feature engineering is not confined to a single operating point but holds across the entire sensitivity-specificity tradeoff space. The separation is most pronounced in the clinically relevant mid-range of false positive rates (approximately 0.2–0.6), which corresponds to the range of thresholds most likely to be used in population-level screening applications where both sensitivity and specificity carry meaningful consequences.

The magnitude of curve separation varies across classifier classes in a way that is itself informative. The elastic net panel shows a consistent and uniform lift, reflecting the fact that the additional features provide linearly separable signal that the regularized logistic model can capture cleanly. The random forest panel exhibits the largest absolute separation, with the enhanced curve pulling furthest from the baseline in the low-to-mid false positive rate region; this gain is consistent with the random forest's capacity to exploit nonlinear interactions between WHtR, the BMI $\times$ sex term, and the race/ethnicity indicators, which the baseline feature set does not provide sufficient structure to leverage. The LightGBM panel shows the smallest separation of the three, suggesting that gradient boosting's capacity to construct complex feature interactions from raw inputs partially compensates for the less informative baseline feature set, though it still cannot recover the full gain available from explicit feature engineering. Together, the ROC panels illustrate that the enhanced feature set uniformly improves discrimination regardless of model class, while also revealing that the mechanism of improvement differs across algorithms in ways consistent with each model's inductive bias.

D. Feature Importance (SHAP Analysis)

SHAP values computed for the best-performing model (enhanced elastic net logistic regression) reveal the contribution of each feature to MetS prediction (Fig. 2). Waist-to-height ratio

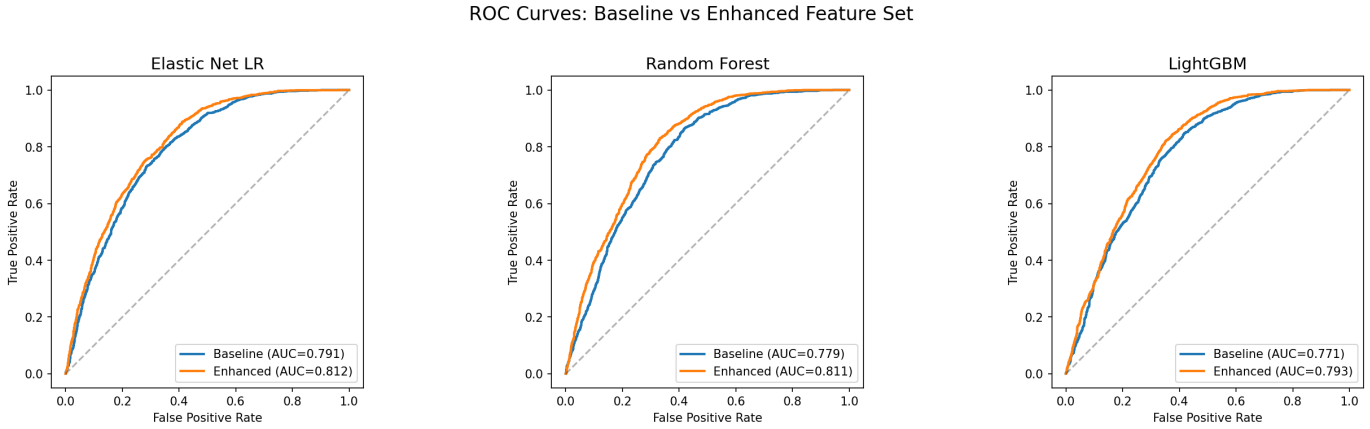


Fig. 1. ROC curves comparing baseline (blue) and enhanced (orange) feature sets across all three classifiers. The enhanced set consistently improves discrimination.

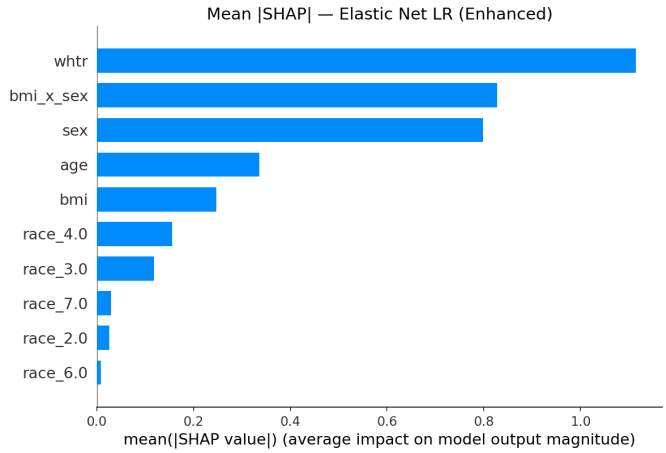


Fig. 2. Mean absolute SHAP values for the enhanced elastic net logistic regression model. WHtR dominates, followed by the BMI × sex interaction.

(WHtR) is the most important predictor by a wide margin, with a mean absolute SHAP value approximately three times that of the next feature. Higher WHtR values push predictions strongly toward MetS, consistent with clinical evidence that central adiposity is a stronger metabolic risk factor than total body weight.

The BMI × sex interaction term ranks second, confirming that allowing the model to learn sex-specific BMI gradients adds substantial predictive value. Sex alone ranks third, reflecting its role in sex-differentiated MetS thresholds. Notably, raw BMI ranks fifth in the enhanced model. Once WHtR and the interaction term are available, BMI alone contributes comparatively little additional information. Race/ethnicity indicators contribute modestly, with the largest effects for Non-Hispanic Black and Non-Hispanic White categories.

E. SHAP Beeswarm Plot

The beeswarm plot in Appendix D provides participant-level detail. For WHtR, the distribution of dots extends far to the right for high-value participants, indicating that elevated waist-to-height ratio consistently and strongly pushes predictions toward MetS across the sample. The low amount of leftward (protective) SHAP contributions for WHtR is itself informative. Unlike BMI, which shows a more symmetric spread around zero, WHtR functions almost unidirectionally in this model, flagging risk. The BMI × sex interaction term shows a similarly directional pattern, with high interaction values clustering on the positive side, confirming that the sex-specific BMI gradient is a consistent source of predictive signal rather than an artifact of a small subset of participants. By contrast, raw BMI shows a compressed distribution centered near zero, reinforcing the conclusion that once WHtR and the interaction term are present, BMI contributes relatively little additional discriminatory information at the individual level.

F. Subgroup Analysis

To evaluate whether the enhanced model performs equitably across demographic groups, we report AUROC, recall, and miss rate by sex and race/ethnicity (Fig. 5, Table IV). The model achieves higher AUROC for males (0.827) than females (0.794), despite higher MetS prevalence among females (38.7% vs 33.6%). Across race/ethnicity groups, performance is generally consistent (AUROC 0.80–0.81 for most groups), with one notable exception: Non-Hispanic Asian participants show lower AUROC (0.793) and a miss rate of 72.6%—meaning the model fails to flag nearly three in four MetS-positive Asian participants at the default threshold, compared to 49.0% for NH Black and 41.5% for NH White participants.

The SHAP heatmap (Fig. 3) and WHtR distribution analysis (Fig. 4) together explain this gap. WHtR carries similar mean absolute SHAP importance across all racial groups (range 0.751–0.986), indicating the model weights it consistently regardless of group. The disparity arises instead because MetS-positive NH Asian participants cluster near or below the WHtR = 0.5 threshold—their MetS criteria are more often met through elevated glucose or adverse lipid profiles than through abdominal adiposity. The model therefore inherits the

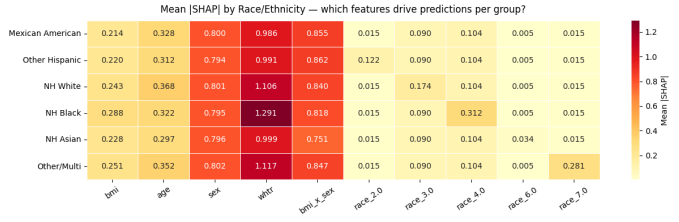


Fig. 3. Mean |SHAP| by race/ethnicity. WHtR importance is consistent across groups (0.751–0.986), indicating the recall gap is a threshold problem rather than a feature weighting problem.

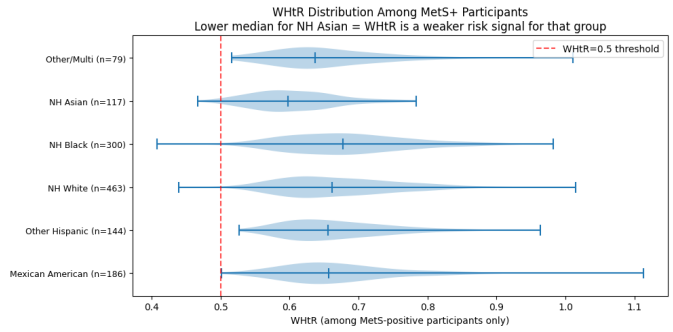


Fig. 4. WHtR distribution among MetS-positive participants by race/ethnicity. NH Asian participants cluster near the 0.5 threshold, consistent with MetS driven by glucose and lipid criteria rather than abdominal adiposity.

insensitivity of standard anthropometric thresholds rather than correcting it.

V. Discussion

Before examining model-level differences, it is worth situating these results within the broader landscape of machine learning approaches to MetS prediction. An AUROC above 0.80 is generally considered strong discrimination for a binary clinical outcome, and achieving this threshold using only anthropometric and demographic features - height, weight, waist circumference, age, sex, and race/ethnicity - is notable. Many competing approaches in the literature rely on laboratory biomarkers such as fasting insulin, liver enzymes, or inflammatory markers that require blood draws and clinical processing. That our enhanced model reaches 0.812 without any such measurements suggests that body-based features, when carefully engineered, carry more predictive information than their simplicity might imply. This does not mean lab-based models are unnecessary, for individuals already presenting with clinical symptoms, richer biomarker data will almost certainly improve prediction further, but it does indicate that a well-constructed anthropometric screening tool could meaningfully identify at-risk individuals earlier and in lower-resource settings where laboratory testing is not immediately available.

Our results demonstrate that a small set of engineered anthropometric and demographic features significantly improves MetS prediction beyond BMI alone and that this improvement is consistent across model classes of varying complexity. Three findings merit particular discussion.

Feature engineering outperforms model complexity. The most striking result is that enhanced elastic net logistic regression (AUROC = 0.812) outperforms both baseline random forest (0.779) and baseline LightGBM (0.771). This suggests that in the context of MetS screening, investing in better feature representation—specifically, features that capture fat distribution and sex-specific risk—yields greater returns than moving to more flexible algorithms. This has practical implications: a simple, interpretable linear model with well-chosen features may be more suitable for clinical deployment than a complex black-box model trained on raw BMI.

WHtR and the BMI $\times$ sex interaction drive the improvement. SHAP analysis confirms that these two engineered features are the primary contributors to the enhanced model's performance. WHtR captures abdominal adiposity, which BMI fundamentally cannot represent because BMI treats all weight equally regardless of where it is distributed. The BMI $\times$ sex interaction allows the model to learn that the same BMI value carries different metabolic risk for men and women—a relationship that the baseline model, which receives BMI and sex as independent inputs, cannot express. These findings align with Yoo (2016), who showed WHtR outperforms BMI for cardiometabolic screening [6], and with Behkamal et al. (2025), who identified sex as a key predictor of MetS [7].

The performance gap between male and female subgroups warrants closer attention. The model achieves an AUROC of 0.827 for male participants and 0.794 for female participants, a difference of 0.033, notable given that females in the sample have a higher MetS prevalence (38.7% versus 33.6%). A higher prevalence typically makes classification easier rather than harder which makes the lower female AUROC more striking. One interpretation is that the BMI $\times$ sex interaction term, while a meaningful improvement over treating sex and BMI as fully independent inputs, captures male metabolic risk gradients more cleanly than female ones. Women's cardiometabolic risk is shaped by factors including hormonal status, reproductive history, and menopausal transition, none of which are represented in the current feature set. The interaction term encodes a static sex-specific BMI slope but it cannot account for the within-sex heterogeneity that these biological and life-course factors introduce. This does not diminish the value of the interaction term as an improvement over the baseline but it does suggest a ceiling on how much a single multiplicative term can accomplish. Future work incorporating hormonal or reproductive covariates, or estimating sex-stratified models independently rather than through an interaction term, may be necessary to close this gap.

The Asian recall gap is a threshold problem, not a feature weighting problem. The NH Asian miss rate of 72.6% is the most consequential equity finding in this analysis. The SHAP heatmap confirms that WHtR is weighted consistently across all racial groups (mean |SHAP| range 0.751–0.986) meaning the model is not deprioritizing this feature for Asian participants. The gap arises because MetS-positive NH Asian

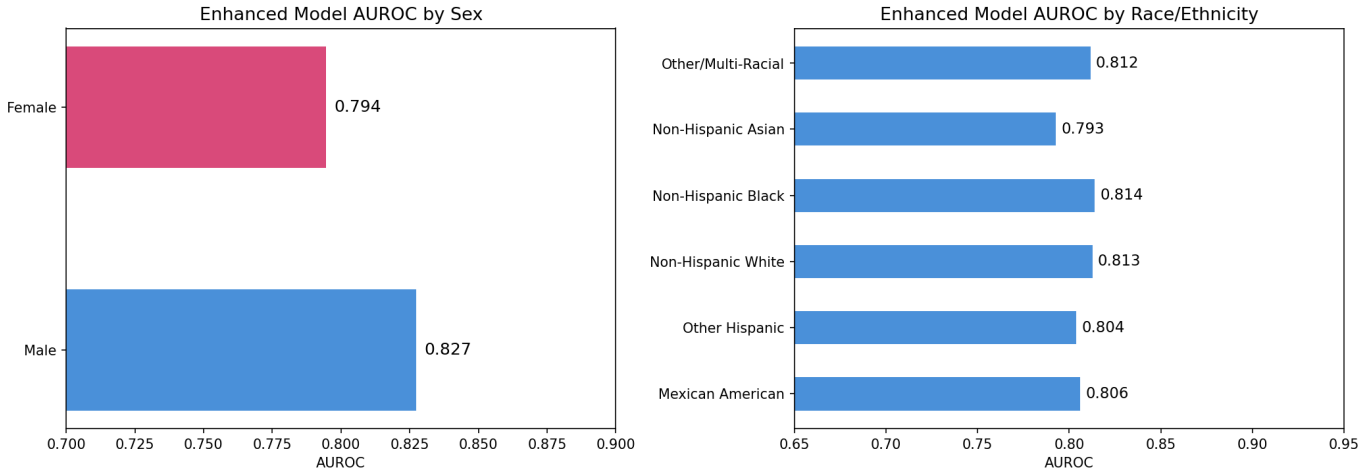


Fig. 5. AUROC of the enhanced elastic net logistic regression model by sex (left) and race/ethnicity (right). Performance is generally equitable, with lower recall for Non-Hispanic Asian participants.

participants tend to present at WHtR values that fall near or below the 0.5 threshold the model associates with lower risk globally. Their MetS is driven by glucose and lipid criteria rather than abdominal adiposity. Lowering the effective WHtR threshold for this group, either through a WHtR $\times$ race/ethnicity interaction term or population-specific cut-offs near 0.47–0.49, is therefore the most targeted intervention available within our current feature framework.

Equity implications. The model learns population-level risk patterns that reflect existing demographic disparities without interrogating the structural and social determinants that produce them. Observed differences in metabolic risk across racial and ethnic groups are shaped by factors including residential segregation, differential access to healthcare, occupational exposure, and chronic stress from systemic discrimination - none of which are captured when race is encoded as a categorical predictor. Any clinical deployment should be accompanied by explicit acknowledgment that race and ethnicity function here as social proxies and that improving model performance for underserved groups requires investment in both better data and more structurally informed features.

We note that WHtR is derived from waist circumference, which is itself a component of the MetS diagnostic criteria. While we deliberately excluded raw waist circumference to prevent direct label leakage, the transformation to a ratio does not fully eliminate information overlap with the outcome. However, WHtR cannot be reverse-engineered to recover the sex-specific waist circumference thresholds used in MetS classification without also knowing the participant's height, and it captures a clinically meaningful construct (central adiposity relative to body frame) that is distinct from the binary diagnostic criterion. We view this as an acceptable tradeoff given our goal of evaluating accessible anthropometric predictors for clinical screening.

Compared to Behkamal et al. (2025), who achieved an AUROC of 0.72 using laboratory biomarkers, our enhanced model achieves 0.812 using only anthropometric and demographic features that require no blood work. This suggests that body-based measures, when appropriately engineered, can match or exceed the predictive value of lab-based approaches for MetS screening, with the added advantage of lower cost and greater accessibility.

Precision-Specificity Tradeoff. A closer examination of the precision and specificity metrics reported in Appendix A reveals a pattern that carries practical clinical significance. The enhanced elastic net model improves precision from 0.644 to 0.665 while maintaining nearly identical specificity (0.855 to 0.847), a difference of less than one percentage point. In a screening context, this tradeoff is meaningful.

Specificity measures the model's ability to correctly identify individuals without MetS, and its near-preservation across feature sets indicates that the enhanced model's improved sensitivity does not come at the cost of substantially more false positives. Put differently, the engineered features help the model find more true cases without proportionally increasing the number of healthy individuals it misclassifies as at-risk. For a screening tool intended for use in clinical or community settings, where unnecessary follow-up testing carries both financial and psychological costs, this stability in specificity alongside gains in precision represents a practically useful, not merely statistical, improvement.

Future Work. Several directions for future work emerge from our findings. First, population-specific WHtR thresholds for Asian participants—prior work suggests a lower cut-off near 0.47–0.49—could substantially improve recall for this group, either through engineered thresholds or a WHtR $\times$ race/ethnicity interaction term. Second, female-specific covariates such as menopausal status or reproductive history may close the male–female AUROC gap that the BMI $\times$ sex term alone cannot resolve. Third, a direct comparison combining our anthropometric features with biomarker data (glucose, lipids, hs-CRP) would clarify whether the two fea-

ture types are complementary or redundant and quantify the accuracy–accessibility tradeoff. Finally, calibration analysis via Brier score and reliability diagrams should accompany any deployment evaluation, particularly for the Non-Hispanic Asian subgroup where threshold-level miscalibration is most likely.

VI. Conclusion

This study evaluates whether a set of engineered body composition and demographic features can improve prediction of metabolic syndrome (MetS) beyond BMI alone. Using NHANES 2017–March 2020 data and a sample of 3,568 non-pregnant adults, we compare a baseline feature set of BMI, age, and sex against an enhanced set that adds waist-to-height ratio, a BMI $\times$ sex interaction term, and race/ethnicity indicators, evaluated across elastic net logistic regression, random forest, and LightGBM classifiers.

The results consistently support our hypothesis. The enhanced feature set significantly improves AUROC across all three classifiers ($p < 0.001$), and the best-performing model, enhanced elastic net logistic regression, achieves an AUROC of 0.812, outperforming all baseline models regardless of their algorithmic complexity. SHAP analysis confirms that WHtR and the BMI $\times$ sex interaction term drive these gains, while raw BMI contributes comparatively little once these engineered features are available. This finding directly addresses a core limitation of BMI identified throughout the literature, its failure to account for fat distribution and sex-specific metabolic risk gradients.

The simplest model in our comparison, when given better features, outperforms more complex tree-based models trained on raw BMI. This supports the argument that BMI's continued dominance in clinical screening reflects historical use and accessibility rather than predictive superiority. When features that capture where fat is distributed on an individual's body are included, the limitations of BMI become quantifiably apparent.

Subgroup analysis introduces an important caveat. While performance is largely consistent across most race/ethnicity groups, the model underperforms for Non-Hispanic Asian participants, producing notably lower recall (0.265) despite an AUROC of 0.793—meaning the model fails to flag approximately 73% of MetS-positive Asian participants at the default threshold. This failure is mechanistic: WHtR, the model's dominant predictor, carries less signal for this group because Asian individuals who meet MetS criteria tend to do so through elevated glucose or adverse lipid profiles rather than high waist-to-height ratio, a pattern consistent with well-documented evidence that standard anthropometric thresholds systematically underestimate cardiometabolic risk in Asian populations. The current feature set, though a meaningful improvement over raw BMI, inherits this insensitivity. Resolving it requires population-specific WHtR thresholds, a WHtR $\times$ race/ethnicity interaction term, or body composition features calibrated to diverse populations—directions that are tractable within the existing NHANES framework and represent the most consequential next step for improving this model's equity.

Taken together, these findings reinforce that a single ratio, derived from weight and height alone, is an insufficient and often inequitable indicator for MetS. The accessibility of BMI can be maintained due to the marginal cost of computing WHtR, requiring only a waist measurement alongside the height already used to calculate BMI, and its beneficial predictive return. Clinicians, researchers, and health organizations may find that incorporating WHtR and sex-aware features alongside BMI offers a more complete picture of metabolic risk at minimal additional cost, a small methodological shift with meaningful implications for screening accuracy and equity.

Contributions

- **Ria:** Introduction, Anti-BMI Literature, Discussion
- **Sachi:** Methodology, Alternative Measures Literature, Dataset, Feature Engineering
- **Diya:** Pro-BMI Literature, Conclusion
- **All:** Abstract, Related Work, Modeling, Results, Bibliography

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Appendix A
Additional Evaluation Metrics

Table II reports precision and specificity for all six model–feature set combinations.

TABLE II
FULL CLASSIFICATION METRICS (MEAN ± SD, 5-FOLD STRATIFIED CV).

Set	Model	Precision	Specificity
Base	EN-LR	.644±.021	.855±.016
Base	RF	.597±.012	.776±.014
Base	LGBM	.585±.013	.775±.016
Enh	EN-LR	.665±.011	.847±.008
Enh	RF	.622±.010	.783±.008
Enh	LGBM	.614±.009	.785±.007

Appendix B
Subgroup Performance Detail

Table III reports per-group metrics for the enhanced elastic net logistic regression model.

TABLE III
SUBGROUP PERFORMANCE OF THE ENHANCED ELASTIC NET LR MODEL.

Group	n	MetS %	AUROC	F1
Male	1,787	33.6%	.827	.579
Female	1,781	38.7%	.794	.604
Mexican American	450	41.3%	.806	.632
Other Hispanic	358	40.2%	.804	.619
Non-Hispanic White	1,238	37.4%	.813	.629
Non-Hispanic Black	901	33.3%	.814	.566
Non-Hispanic Asian	428	27.3%	.793	.354
Other/Multi-Racial	193	40.9%	.812	.625

TABLE IV
MISS RATE (FALSE NEGATIVE RATE) AMONG METS-POSITIVE PARTICIPANTS BY RACE/ETHNICITY, ENHANCED ELASTIC NET LR MODEL.

Group	MetS+ n	Recall	Miss Rate
NH Asian	117	0.274	0.726
NH Black	300	0.510	0.490
NH White	463	0.585	0.415
Other Hispanic	144	0.590	0.410
Mexican American	186	0.602	0.398
Other/Multi	79	0.620	0.380

Appendix C
Precision-Recall Analysis

Fig. 6 shows precision-recall curves for all six model–feature set combinations. Average Precision (AP) improves from the baseline to the enhanced set across all classifiers: elastic net LR (0.622 to 0.654), random forest (0.603 to 0.644), and LightGBM (0.586 to 0.615).

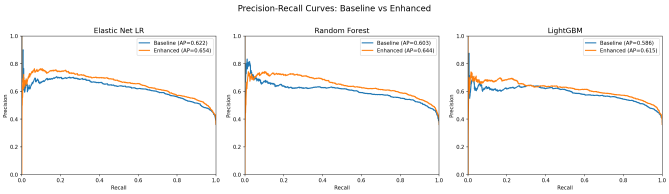


Fig. 6. Precision-recall curves for all six model–feature set combinations.

Appendix D
SHAP Summary Plot

Fig. 7 shows the detailed SHAP summary (beeswarm) plot. Each point represents a single participant. Color indicates feature value (red = high, blue = low) and horizontal position indicates the SHAP contribution to the predicted MetS probability.

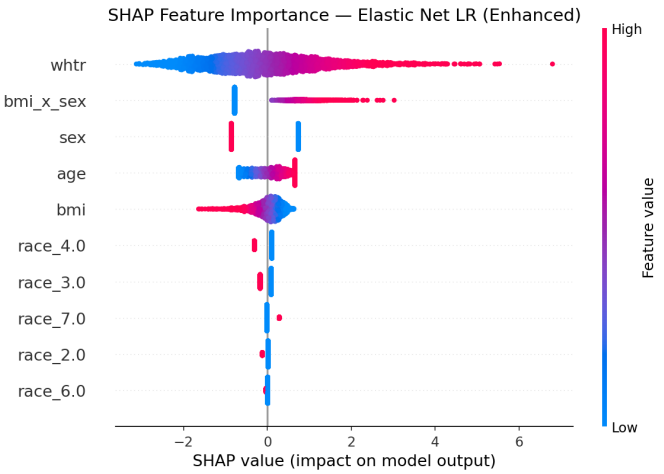


Fig. 7. SHAP beeswarm plot for the enhanced elastic net logistic regression model. Each dot is one participant; color indicates feature value.